

CASE 22

WARTS, HYPOGAMMAGLOBULINEMIA, INFECTIONS, AND MYELOKATHEXIS SYNDROME (WHIM SYNDROME)

Chemokines direct the migration of leukocytes to where they are needed.

Leukocytes generated in the bone marrow are released into the bloodstream, from which they populate peripheral lymphoid organs and patrol the periphery to detect invading pathogens. Effector leukocytes circulate in the bloodstream and can be specifically recruited to sites of infection when required. Naive lymphocytes continuously recirculate from the bloodstream into peripheral lymphoid organs and back to the bloodstream (Fig. 22.1). Leukocyte migration, or 'homing,' to lymphoid organs and infected tissues is controlled mainly through interactions between chemokines and chemokine receptors.

Chemokines produced at sites of inflammation and/or infection attract leukocytes that express specific receptors for that chemokine, which promotes their extravasation. In contrast, chemokines constitutively expressed by stromal cells within the bone marrow and lymphoid organs contribute to immune homeostasis and maintain the architecture of lymphoid tissues.

The chemokine CXCL12 [stromal derived factor 1 (SDF-1)] is a key chemokine that directs the homing of both precursor and mature leukocytes to the bone marrow. It binds to the receptor CXCR4 expressed by leukocytes. Levels of CXCR4 on the surface of leukocytes vary depending on the maturation and activation status of

TOPICS BEARING
ON THIS CASE:

Neutrophil function

Chemokines and
their receptors

Leukocyte homing

Leukocyte
development

This case was prepared by Luigi Notarangelo, MD, in collaboration with Anna Virginia Gullino, MD.

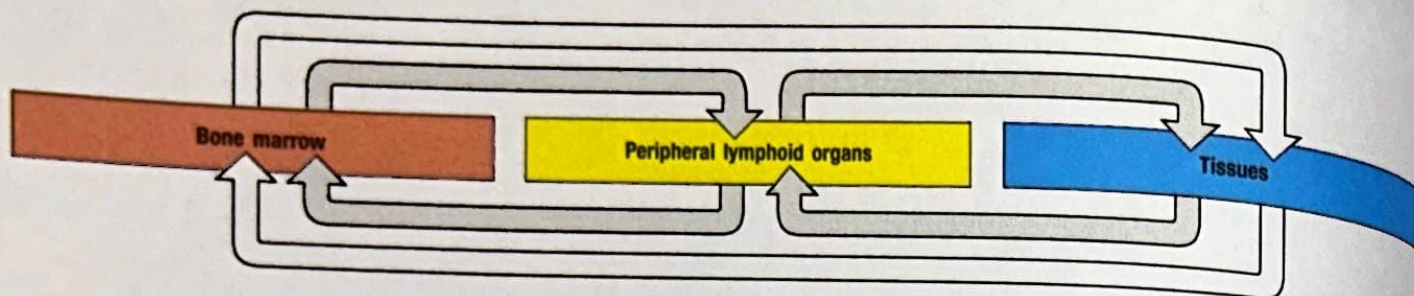


Fig. 22.1 Recirculation of leukocytes between the bone marrow, peripheral lymphoid organs, and peripheral tissues.

the cells, thereby modulating their responsiveness to CXCL12 and their tendency to home to the bone marrow.

In particular, CXCR4 is critical in guiding hematopoietic stem cells to the bone marrow during embryogenesis. In postnatal life, it helps maintain hematopoietic stem cells in the bone marrow stem-cell niche. In leukocytes of the myeloid lineage, CXCR4 has a bimodal pattern of expression; higher levels are present on the surfaces of immature myeloid progenitors and senescent neutrophils than on the surfaces of mature leukocytes. The CXCR4 ligand CXCL12 is produced at high levels by bone marrow stromal cells, such that the interaction of CXCL12 with CXCR4 promotes the retention of immature myeloid progenitors in the bone marrow (permitting their maturation), and also favors the elimination of senescent neutrophils in the periphery by recruiting them to the bone marrow, where they undergo apoptosis. Concurrently with this dual pattern of expression of CXCR4 during their maturation, myeloid cells also show variable levels of expression of the chemokine receptor CXCR2, which also has an important role in modulating the circulation of myeloid cells by its interaction with several ligands, including CXCL1 and CXCL2.

CXCR4 is also important for the recirculation and homing of lymphoid lineage cells; in particular, along with CXCR5 (the receptor for the chemokines CXCL13 and CXCL21), it modulates the migration of B lymphocytes within the follicles and germinal centers of the lymph nodes. CXCR4 also has an important role in T lymphocytes. In particular, it is a co-receptor for HIV-1, which facilitates the infection of CD4⁺ T cells by the virus.

As we shall see in the case described below, mutations in the *CXCR4* gene in humans are associated with an immunodeficiency that results from deficiencies in leukocyte trafficking as a result of defective interaction between the CXCL12 chemokine and its receptor.

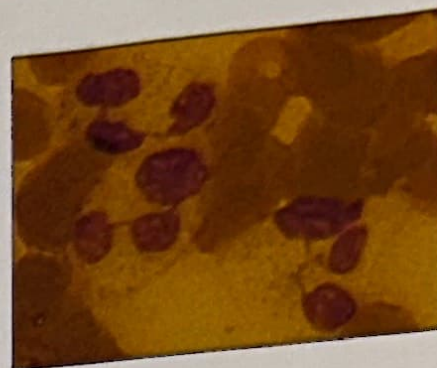
The case of Sue Bridehead: recurrent pyogenic infections with severe chronic neutropenia but hypercellular marrow.

Sue was born with the congenital heart defect Tetralogy of Fallot. Severe neutropenia (absolute neutrophil count (ANC) $350 \mu\text{l}^{-1}$; normal value $3000\text{--}5800 \mu\text{l}^{-1}$) was detected at 2 years of age during work-up for secondary surgical correction of her heart defect, but she was not investigated further. Between 2 and 4 years of age, Sue had two episodes of pneumonia, which were treated with amoxicillin and clavulanic acid by her primary care physician, but were not investigated further. At the age of 4 years, Sue was admitted to Children's Hospital Boston because of a third episode of bacterial pneumonia. She had had a fever (temperature 38.7°C) and a cough for the past 3 days. Rales (crackles) could be heard at the right lower lobe, and a murmur at the left sternal border. She had very mild dyspnea (shortness of breath) with oxygen saturation of 96%. Sue was at the 10th centile for both height and weight and had no dysmorphic features. Her tonsils were small, there was no hepatosplenomegaly, and the rest of her physical examination was unremarkable.

Recurrent pneumonia.

Case 22: Warts, Hypogammaglobulinemia, Infections, and Myelokathexis Syndrome (WHIM Syndrome)

Fig. 22.2 Bone marrow examination in patients with WHIM syndrome reveals accumulation of senescent, hypersegmented neutrophils. Three neutrophils with hypersegmented nuclei (stained red with dark-staining chromatin) can be seen. Staining with May-Grünwald-Giemsa. Photograph courtesy of Antonio Regazzoli and Lucia Notarangelo, Spedali Civili Brescia, Italy. Magnification $\times 40$.



She was the only child of healthy non-consanguineous parents. Other than the bacterial pneumonias, she had not suffered significantly from infectious diseases and had received immunizations according to the normal schedule.

A chest X-ray confirmed right inferior lobar pneumonia. Laboratory investigations showed a normal ANC ($2700 \mu\text{l}^{-1}$) but elevated C-reactive protein (CRP; $17 \text{ mg } \mu\text{l}^{-1}$). Sue was started on antibiotic treatment with amoxicillin and clavulanate, and fever subsided after 48 hours. She was discharged, but a week later she was found to be leukopenic (white blood cell count $1500 \mu\text{l}^{-1}$) with significant neutropenia (ANC $300 \mu\text{l}^{-1}$) and lymphopenia (absolute lymphocyte count (ALC) $1000 \mu\text{l}^{-1}$; normal more than $2000 \mu\text{l}^{-1}$). Hemoglobin levels and platelet count were normal. CRP had also normalized. She was found to be hypogammaglobulinemic, with IgG levels of 225 mg dl^{-1} (normal $345\text{--}1236 \text{ mg dl}^{-1}$), IgA 12 mg dl^{-1} (normal $14\text{--}159 \text{ mg dl}^{-1}$), and IgM 33 mg dl^{-1} (normal $43\text{--}207 \text{ mg dl}^{-1}$). These tests were repeated after a month, and confirmed the presence of leukopenia with severe neutropenia and moderate lymphopenia, along with hypogammaglobulinemia.

In spite of the fact that Sue had received a full course of immunizations, her antibody titers to tetanus toxoid, *Haemophilus influenzae*, and *Streptococcus pneumoniae* were nonprotective. The severe neutropenia prompted her physicians to take a bone marrow aspirate, which showed very active myelopoiesis with myelokathexis—that is, the accumulation of senescent neutrophils characterized by pyknotic and hypersegmented nuclei (Fig. 22.2). The possibility of WHIM syndrome was considered. Sequence analysis of the *CXCR4* gene revealed heterozygosity for the R334X nonsense mutation. Sue was treated with daily human recombinant granulocyte colony-stimulating factor (hrG-CSF, $5 \mu\text{g}$ per kg body weight) and regular administration of intravenous immunoglobulin. She is now 12 years old and has not had any other severe bacterial infection; however, she has recently begun to develop multiple warts, which are refractory to topical treatment and tend to recur after curettage.

Neutropenia and hypogammaglobulinemia.

Aberrant bone marrow morphology, with abundance of senescent neutrophils.

WHIM syndrome.

WHIM syndrome is very rare, with a frequency less than 1 case per million individuals. The gene responsible for WHIM syndrome has been mapped to chromosome 2. It encodes the *CXCR4* protein, a G-protein-coupled chemokine receptor containing seven transmembrane domains (Fig. 22.3), which specifically binds the chemokine CXCL12. Most patients with WHIM syndrome carry a heterozygous mutation that leads to premature truncation of the carboxy-terminal tail (C-tail) of the *CXCR4* protein. Frameshift and missense mutations in the C-tail of *CXCR4* have also been reported, but more rarely. The mutant protein is normally synthesized and expressed at the cell surface. It may form homodimers or heterodimer complexes with the normal *CXCR4* protein (encoded by the wild-type allele). These dimers maintain the ability to deliver intracellular activating signals but are refractory to β -arrestin-dependent endocytosis. Hence, WHIM-causing *CXCR4* mutations result in increased chemokine receptor signaling. As a result, neutrophils and lymphocytes from WHIM patients show an increased chemotactic response to CXCL12.

A small minority of patients are phenocopies. Some of these patients have low levels of the kinase G-protein-coupled receptor kinase 3 (GRK3), which mediates

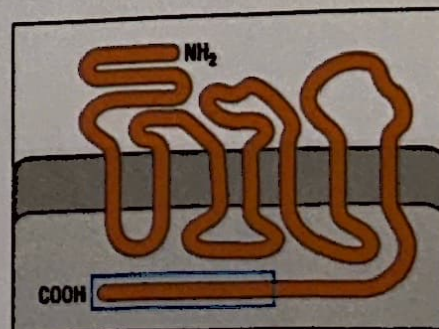


Fig. 22.3 Structure of the chemokine receptor CXCR4, a G-protein-coupled receptor. The structure is shown schematically. Note the seven membrane-spanning domains. The boxed area indicates the carboxy-terminal domain of the protein where heterozygous and frameshift mutations that cause WHIM syndrome are clustered.

CXCL12-dependent internalization and desensitization of CXCR4, and hence acts as a negative regulator of CXCR4 signaling. However, no mutations of the *GRK3* gene have been identified so far in these patients.

Although the acronym WHIM was coined after the description of a series of patients who shared all the typical clinical and laboratory features of the disease (warts, hypogammaglobulinemia, infections, and myelokathexis), after the discovery of the gene defect it has become obvious that patients can also present with an incomplete phenotype. In particular, warts may not become evident for several years after birth.

Peripheral leukopenia, and especially a marked reduction of the absolute neutrophil count, is almost invariably present. These peripheral data contrast with a hypercellularity in the bone marrow due to an expansion of mature and apoptotic neutrophils. The Greek-derived term 'myelokathexis' indicates white blood cell retention. The neutropenia may be somewhat ameliorated during acute infections, when pro-inflammatory signals outweigh the CXCL12-mediated retention of neutrophils in the bone marrow. Furthermore, G-CSF and granulocyte-macrophage colony-stimulating factor (GM-CSF), epinephrine, and glucocorticoids also rapidly stimulate leukocyte release. Daily administration of hrG-CSF is often used in the treatment of patients with WHIM and a history of multiple severe infections. A phase 1 clinical trial has been recently conducted with the use of plerixafor, a CXCR4 antagonist. When administered at low dosage, this drug was able to increase significantly the count of circulating neutrophils in patients with WHIM.

Recurrent bacterial infections are common in WHIM syndrome, and are mostly due to common pathogens such as *H. influenzae*, *Staphylococcus aureus*, and *S. pneumoniae*. Infections often involve the respiratory tract, but deep-seated abscesses can also occur. During the acute phase of infection, the absolute neutrophil count can be normal or even elevated, indicating that WHIM neutrophils may leave the bone marrow in response to strong inflammatory signals.

Hypogammaglobulinemia is frequently present, and the count of circulating B lymphocytes is often decreased. In particular, there is a reduction in the number of switched memory (CD27⁺ IgD⁻) B cells. Patients with WHIM syndrome can mount specific antibody responses after immunization, but the titer of specific antibodies often wanes with time. Immunoglobulin replacement therapy should be considered in patients with hypogammaglobulinemia and is effective in reducing the risk of bacterial infections. Patients with WHIM syndrome also have a reduction in the number of naive T cells, and a relative expansion of effector memory T cells, which often show a restricted repertoire.

Most, but not all, patients with WHIM syndrome develop persistent and treatment-refractory warts (Fig. 22.4), which result from infection by the human papillomavirus (HPV). Typically, the hands are involved, but warts may appear on any part of the body. Treatment with intravenous immunoglobulin or hrG-CSF does not cause regression of the warts. Genital warts (condylomata acuminata) affecting the anogenital tract may predispose to epithelial cancer. For a long time, it was thought that patients with WHIM syndrome are uniquely susceptible to HPV but not to other viruses. However, an increased frequency of infections by herpes simplex virus (HSV), human herpesvirus 8 (HHV8, associated with Kaposi disease), the Epstein-Barr virus (EBV), and the viral skin infection molluscum contagiosum has recently been shown. EBV infection can lead to lymphoma, especially involving the skin (cutaneous lymphoma). Multiple immunological abnormalities may account for the high rate of warts in patients with WHIM syndrome. In particular, the number of circulating plasmacytoid dendritic cells, the major source of interferon- α (IFN- α), is markedly reduced. Furthermore, a defect in the trafficking of effector cells (T cells and NK cells) and antigen-presenting cells (such as dendritic cells) may also contribute to cutaneous viral infections. An increased risk of cutaneous malignancies (T-cell lymphoma and basal cell carcinoma) has been reported.



Fig. 22.4 Typical warts in a patient with WHIM syndrome.

Finally, the incidence of congenital cardiac malformations in patients with WHIM syndrome (7%) is substantially higher than in the general population (0.8%), which is consistent with an established role for CXCR4 in cardiac patterning.

Questions.

- 1 Sue's laboratory features included neutropenia and hypogammaglobulinemia. Is WHIM syndrome the only immunodeficiency that presents with this association? What features might you look for to distinguish WHIM syndrome from other conditions that present with hypogammaglobulinemia and neutropenia?
- 2 What are the most likely reasons that neither of Sue's parents showed any clinical features of WHIM?
- 3 What mechanisms might account for the increased susceptibility to viral cutaneous infections of patients with WHIM syndrome?
- 4 What kind of pharmacological intervention could you suggest that would interfere with the CXCL12-CXCR4 interaction? Can you think of a useful clinical application for such an intervention?